KING SALMAN INTERNATIONAL UNIVERSITY

Field of ENGINEERING

Engineering Program
All Engineering Programs

MAT121 Statics

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***0*** LAB + ***0*** OTH)

Lecture : (Vectors for mechanics)

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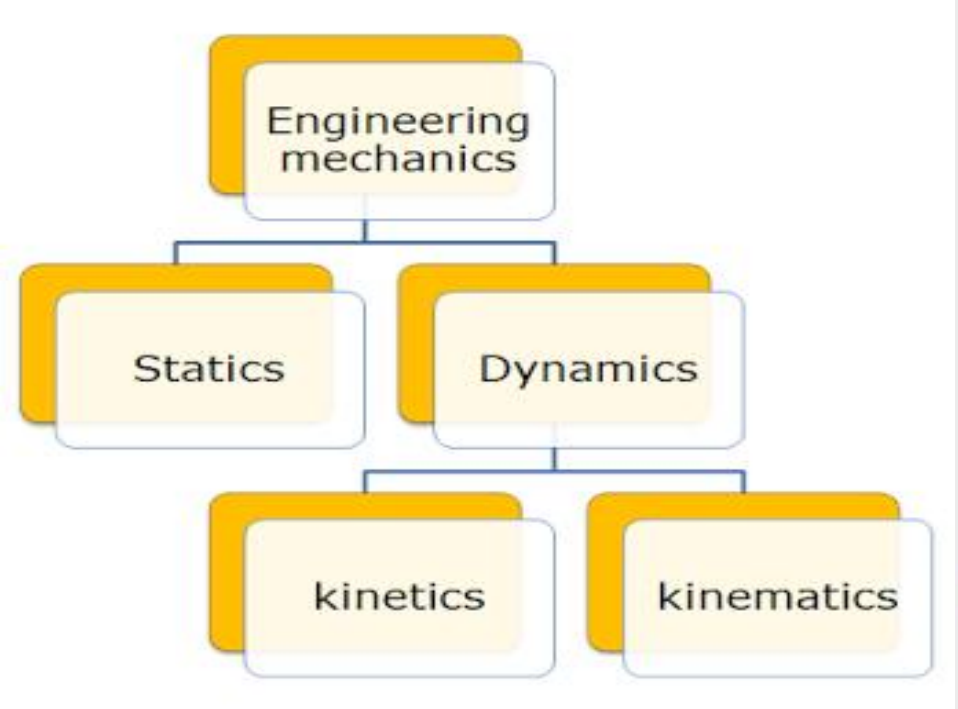
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What is Mechanics?

- Mechanics is the study of bodies under the action of forces.
- Mechanics is an applied science, closely related to physics, so many of the concepts will build on that prior knowledge.
- Mechanics is the foundation of many engineering topics and is an indispensable prerequisite to their study.

Categories of Mechanics:

- Rigid bodies.
  - *Statics* – bodies at rest or at constant velocity.
  - Dynamics – accelerating bodies.
- Deformable bodies.
- Fluids – gas and/or liquid.



Chapter 1
Vectors for mechanics

This chapter introduces **vector skills** in the context of mechanics. Notational clarity is emphasized because correct calculation is impossible without distinguishing fundamental quantities.

The most fundamental quantities in mechanics:

1 - Scalar:

A scalar is any positive or negative physical quantity that can be completely specified by its **magnitude only**.

Mass, time and length are examples on scalar quantity.

2 - Vector:

A vector is any physical quantity that requires **both** a **magnitude** and a **direction** for its complete description.

Force, moment and displacement are examples on vector quantity.

Scalar vs Vector

SCALAR

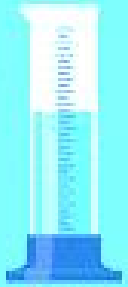
A scalar quantity has magnitude only.



speed



mass



volume



time

VECTOR

A vector has both magnitude and direction.



velocity



weight

friction



Vector Representation:

There are two ways to represent a vector, **graphical** and **analytical** representation

1- Graphical Representation

Every vector in mechanics is well visualized as an arrow, as shown in figure (1 – 1). The direction of the arrow is the direction of the vector. The length of the arrow is proportional to the magnitude of the vector.

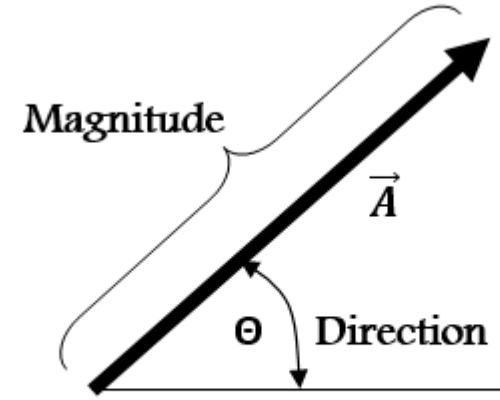


Figure (1 – 1) Graphical Representation of a Vector $\vec{A}$

In order to specify a vector completely the following particulars must be given:

(1) The magnitude

(2) Its direction.

(3) A point on its direction.

The negative of a given vector $\vec{A}$ is a vector of the same magnitude, but opposite direction; this new vector is denoted by $-\vec{A}$ as shown in figure (1 – 2)

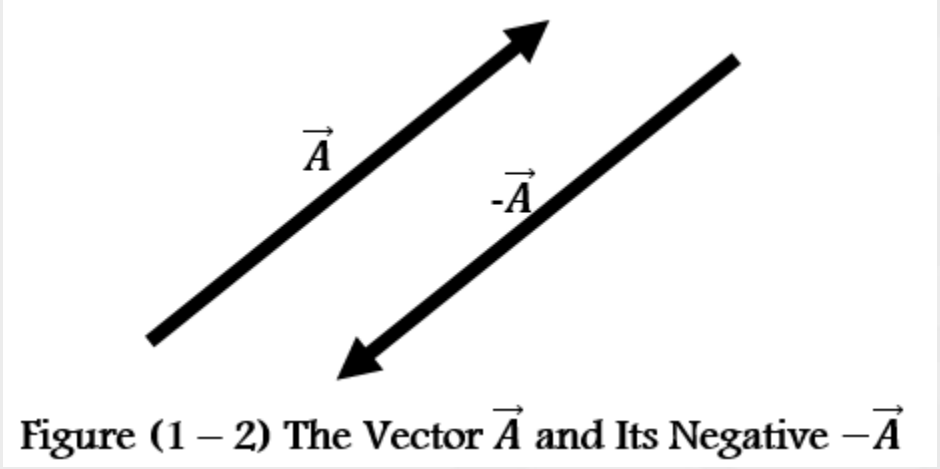


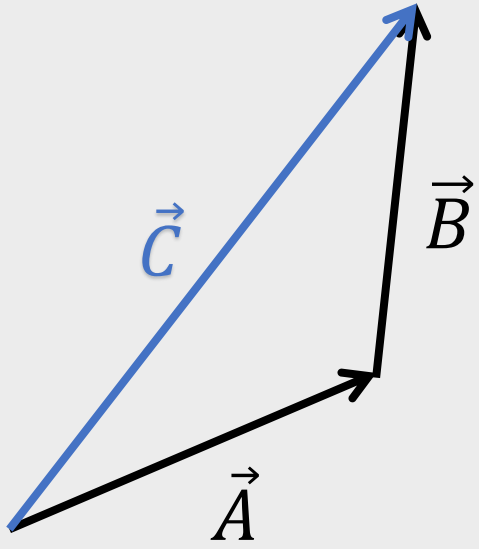
Figure (1 – 2) The Vector $\vec{A}$ and Its Negative $-\vec{A}$

Also vectors mathematical operations (addition and subtraction) can be represented graphically as the following:

Vectors Addition

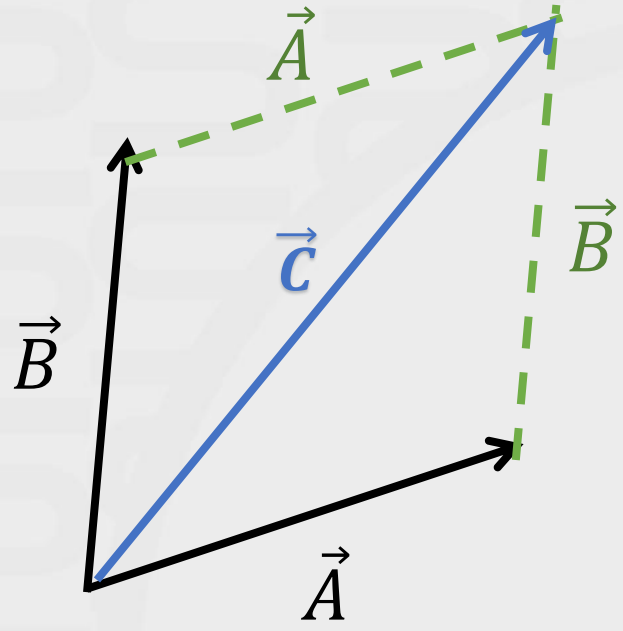
For any two vectors $\vec{A}$ and $\vec{B}$ can be added to obtain a vector $\vec{C}$, which is called ***the resultant.***

Triangle Rule



The tail of $\vec{B}$ on **the head** of $\vec{A}$

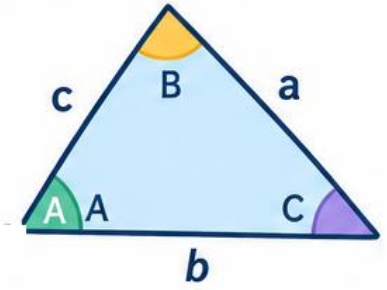
Parallelogram Rule



The tail of $\vec{B}$ on **the tail** of $\vec{A}$

Sine Rule vs Cosine Rule

Sine Rule

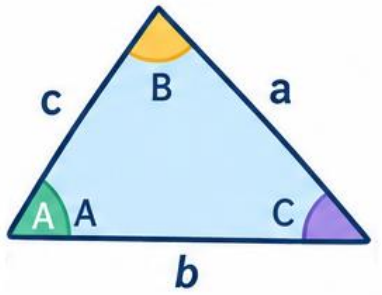


$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

When to use it

- When you know two angles and one side
- When you know two sides and an angle opposite one of them

Cosine Rule



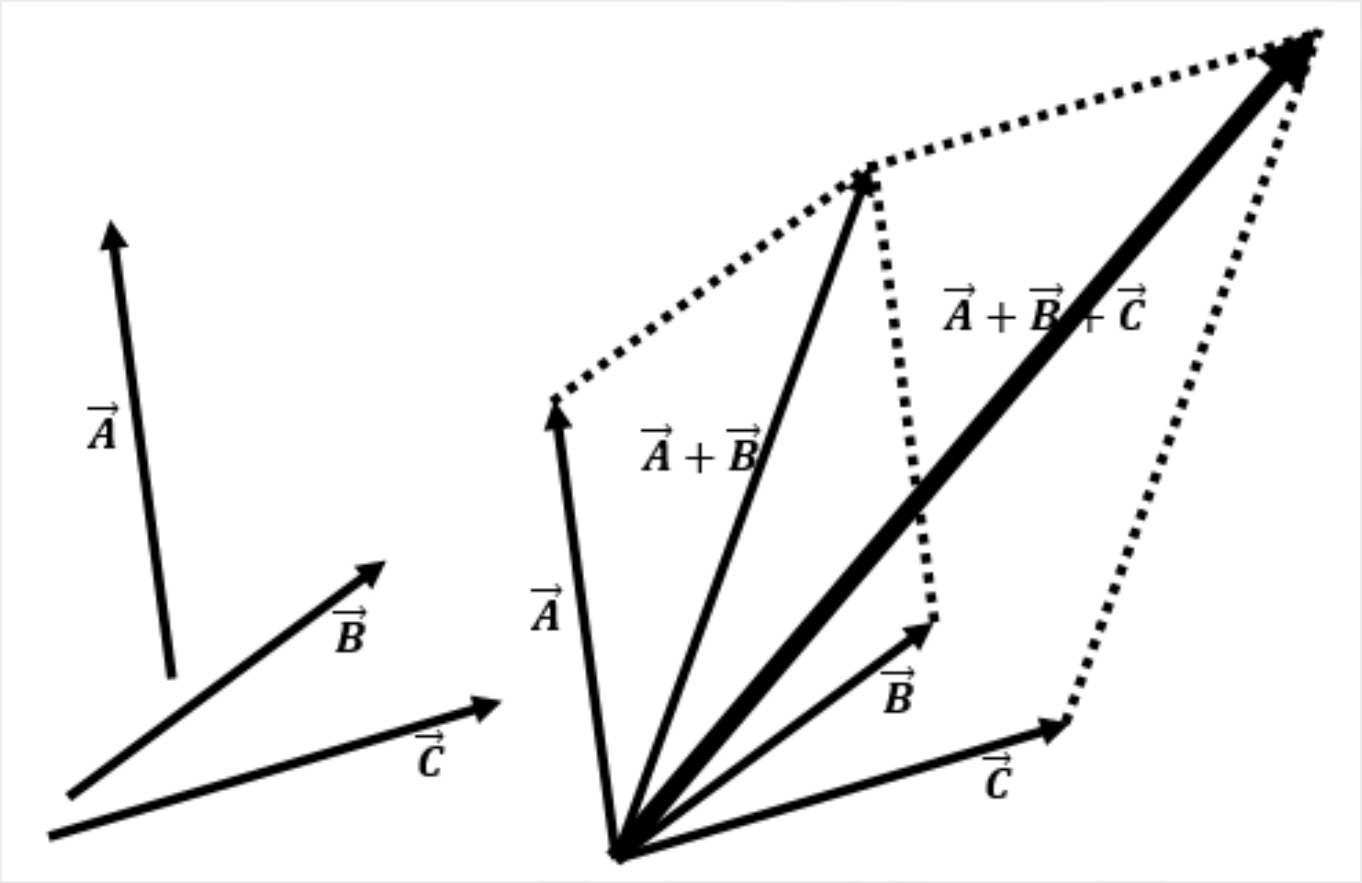
$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

When to use it

- When you know three sides (SSS).
- When you know two sides and the included angle (SAS).

If more than two vectors are to be added. Successive applications of the parallelogram law can be carried out in order to obtain the resultant.

For example, to obtain the resultant of the three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$, as shown in figure (1 – 5), first the resultant of $\vec{A}$ and $\vec{B}$ is found then it is added to $\vec{C}$ to get the resultant of the three vectors.



➢ Vectors Subtraction:

- Where the subtraction of two vectors $\vec{A}$ and $\vec{B}$ is defined as the sum of $\vec{A}$ and$-\vec{B}$, so vectors subtraction can be executed using parallelogram rule.
- Figure (1 – 6) shows the vectors subtraction obtained by adding $\vec{A}$ and $-\vec{B}$ using parallelogram rule.

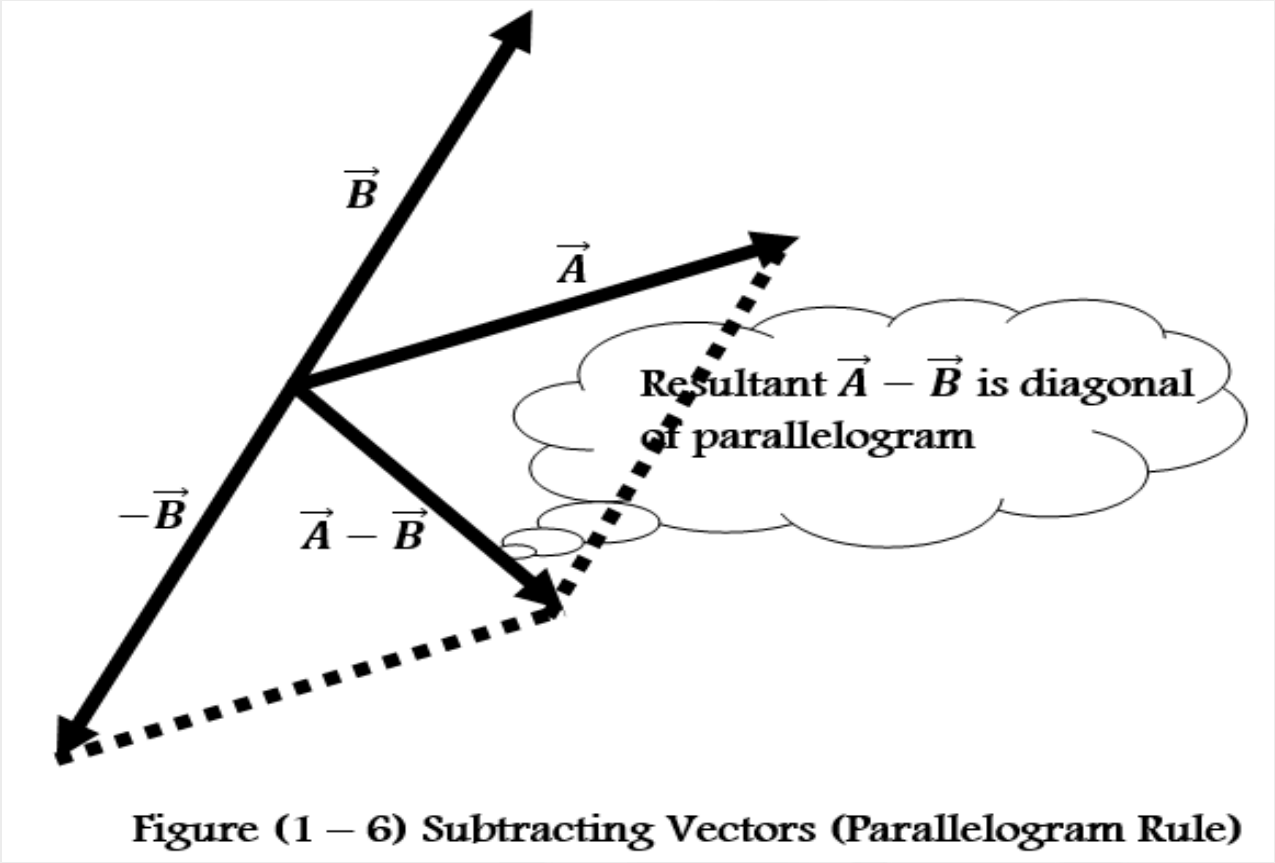
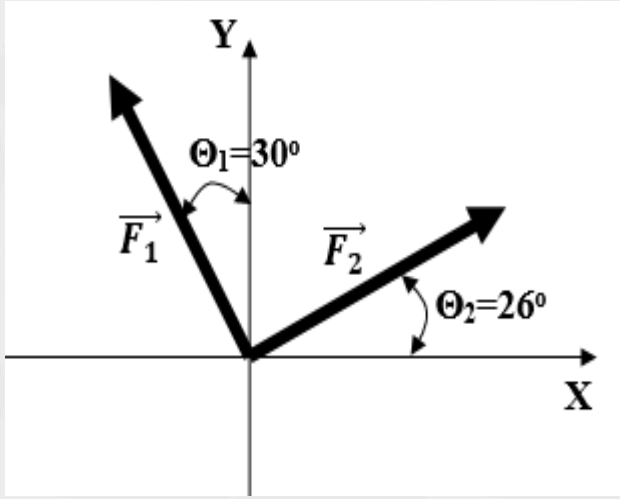
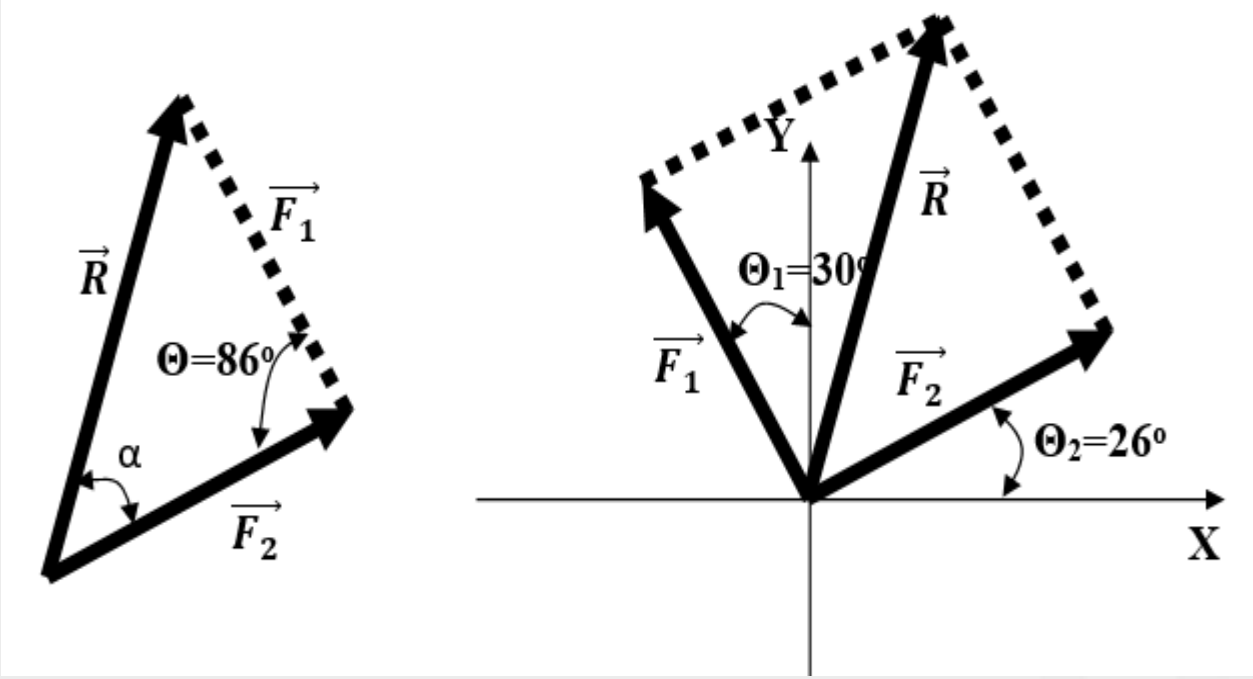


Figure (1 – 6) Subtracting Vectors (Parallelogram Rule)

Example 1 – 1:

Using parallelogram rule, get the magnitude of the resultant of the two forces $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$ which have magnitudes 50 N and 75 N respectively as shown in the figure. Also determine the direction of the resultant measured counter clockwise from the positive direction of x axis.





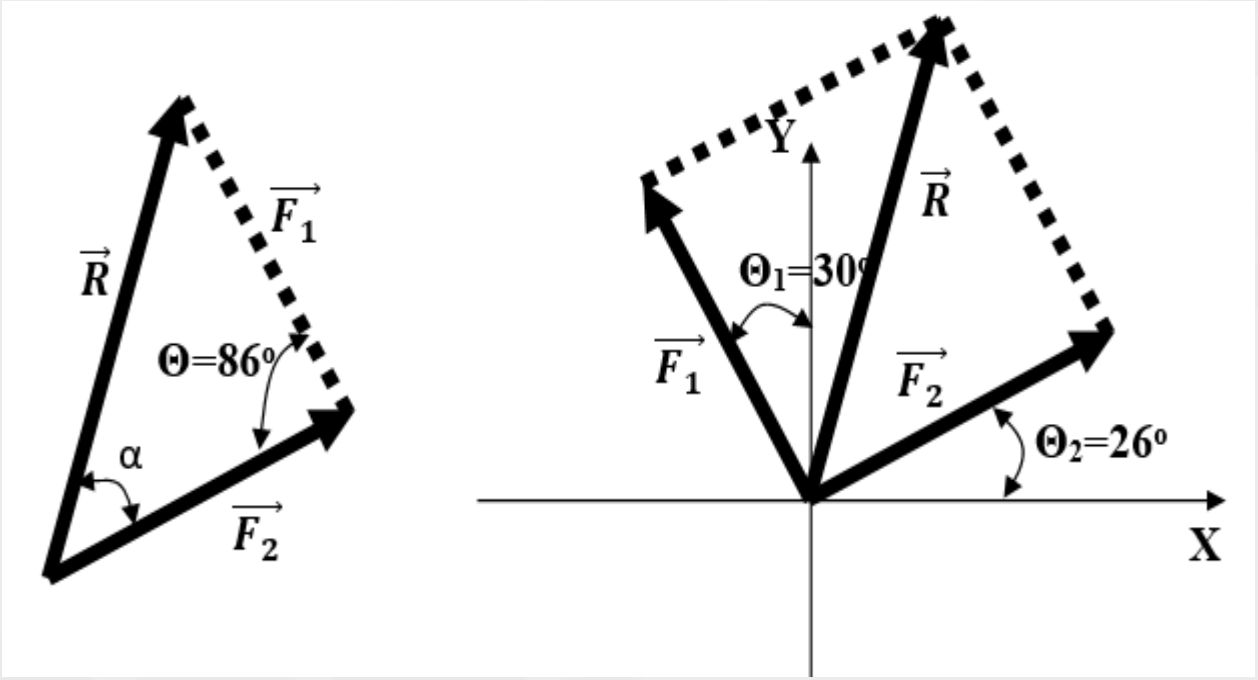
- Draw parallelogram as shown in the following figure. The diagonal of the parallelogram is the resultant $\vec{R}$.
- Take right or left triangle of the parallelogram as shown.
- Apply trigonometry to obtain the magnitude and direction of the resultant

From cosine rule

$$\because R = \sqrt{F_1^2 + F_2^2 - 2F_1F_2\cos\theta}$$

$$\therefore R = \sqrt{50^2 + 75^2 - 2 \times 50 \times 75 \cos 86}$$

$$\therefore R = 87.188 \text{ N}$$



And by using sine rule, then:

$$\because \frac{R}{\sin 86} = \frac{F_1}{\sin \alpha}$$

$$\therefore \frac{87.188}{\sin 86} = \frac{50}{\sin \alpha}$$

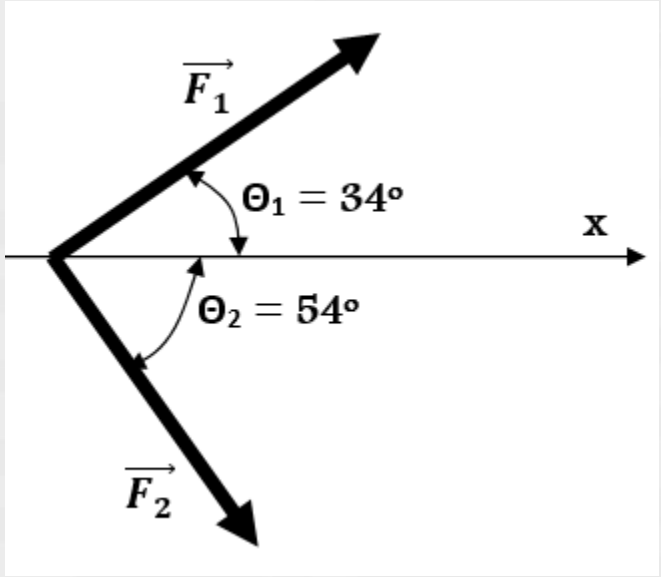
$$\therefore \sin \alpha = 0.572$$

$$\therefore \alpha = 34.895^o$$

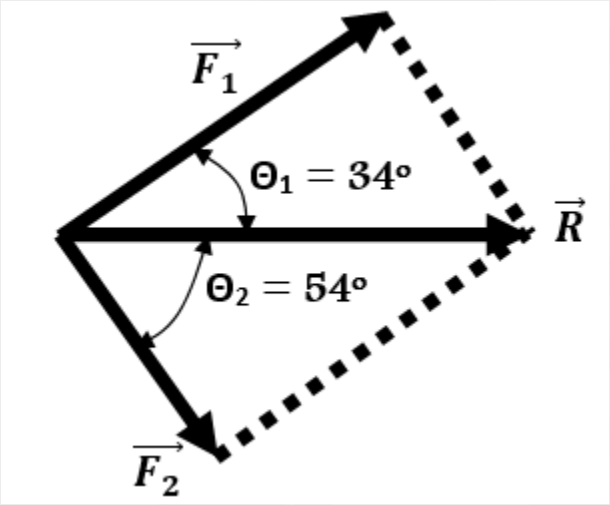
∴The Direction of the resultant measured from the positive direction of the positive x axis = 34.895 + 26 =60.895°

Example 1 – 2:

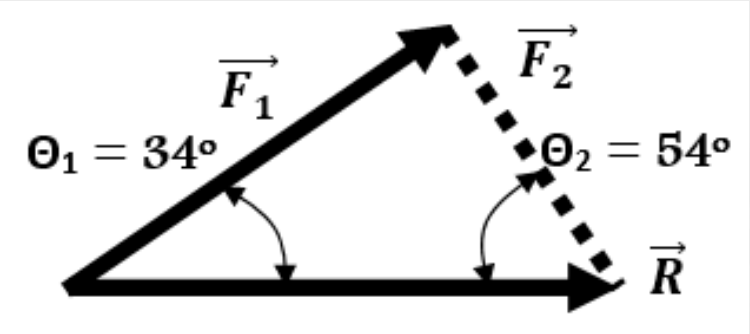
For the shown two forces $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$, where $\overrightarrow{F_1}$ has a magnitude of 50 N. Determine the magnitudes of the force $\overrightarrow{F_2}$ and its resultant so that the resultant is directed in positive direction of x axis.



- Draw parallelogram as shown in the following figure. The diagonal of the parallelogram $\vec{R}$ coincide with x axis.



- Take upper or lower triangle of the parallelogram as shown.



- Apply trigonometry to obtain the magnitudes of the resultant and $\overrightarrow{F_2}$

From sine rule:

$$\because \frac{F_1}{\sin 54} = \frac{F_2}{\sin 34} = \frac{R}{\sin(180 - (54 + 34))}$$

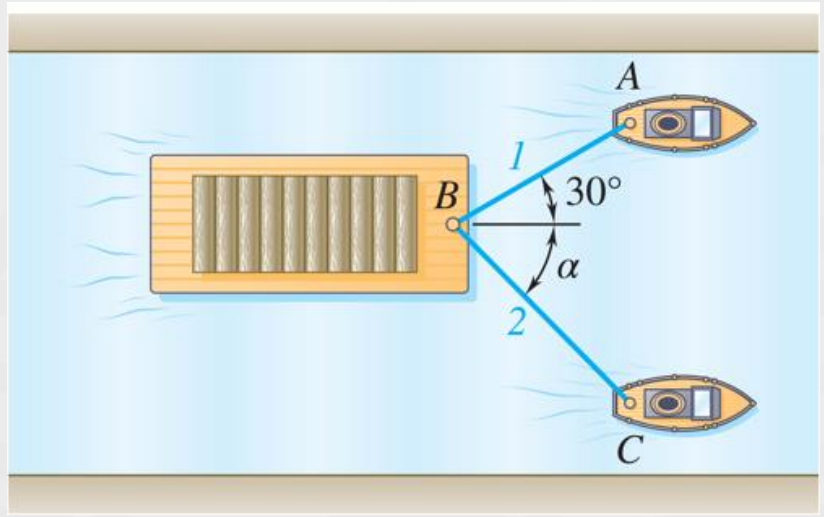
$$\therefore \frac{50}{\sin 54} = \frac{F_2}{\sin 34} = \frac{R}{\sin(92)}$$

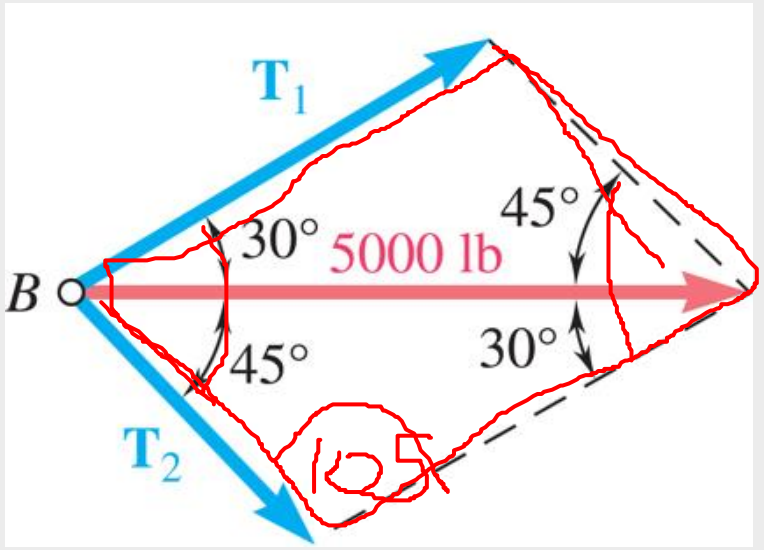
$$\therefore F_2 = \frac{50}{\sin 54} \times \sin(34) = 34.56\ N$$

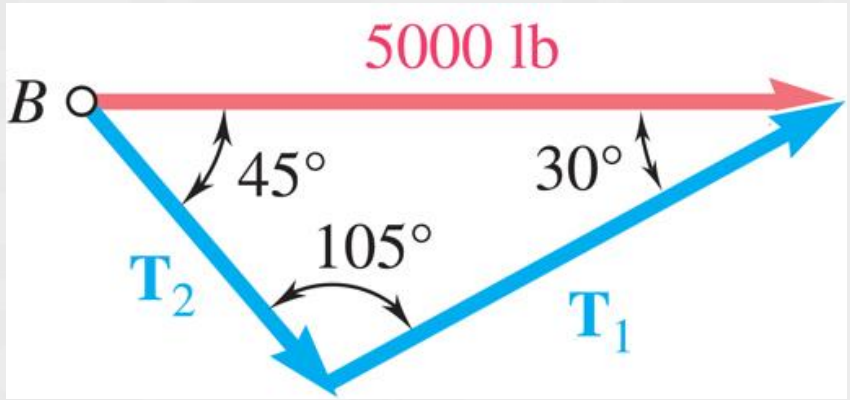
$$\therefore R = \frac{50}{\sin 54} \times \sin(92) = 61.765\ N$$

Example 1 – 3:

A barge is pulled by two tugboats. If the resultant of the forces exerted by the tugboats is 5000 lb directed along the axis of the barge, determine the tension in each of the ropes when **α = 45°**.







$$\frac{T_1}{\sin 45^\circ} = \frac{T_2}{\sin 30^\circ} = \frac{5000\text{lb}}{\sin 105^\circ}$$

$$\boxed{T_1 = 3660\text{lb} \quad T_2 = 2590\text{lb}}$$

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